

Experiment No. 32

Title

Prolog Program to Implement Pattern Matching

Aim

To write a Prolog program to implement **pattern matching**.

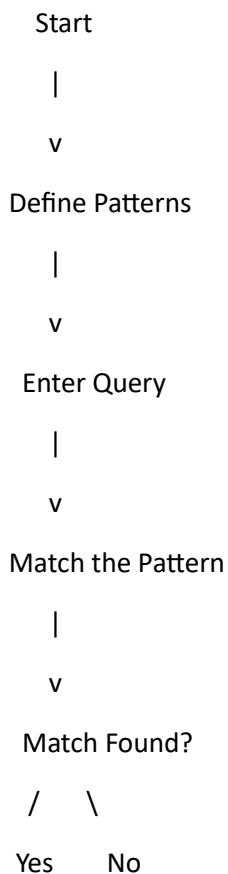
Objective

The objective of this experiment is to understand how Prolog matches a given pattern with available facts and retrieves the matching result.

Algorithm

1. Start.
2. Define patterns and their corresponding values.
3. Enter a pattern as a query.
4. Compare the query with the stored patterns.
5. Display the matching result.
6. Stop.

Flowchart



| |
v v

Display Match Display

No Match

\ /
v v

Stop

Prolog Program

pattern(hello, greeting).

pattern(bye, farewell).

pattern(hi, greeting).

pattern(thanks, gratitude).

match(Pattern, Meaning) :-

 pattern(Pattern, Meaning).

Queries

?- match(hello, Meaning).

Sample Output

Meaning = greeting.

Another query:

?- match(bye, Meaning).

Output:

Meaning = farewell.

Result

The Prolog program successfully implemented **pattern matching** and retrieved the corresponding meaning for the given pattern.

Conclusion

Thus, pattern matching was successfully implemented in Prolog using facts and queries.